



Cloud Computing Lab

Lab #1

Title : GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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Registration number: FA24B1-SE-050

Course code: SE-665

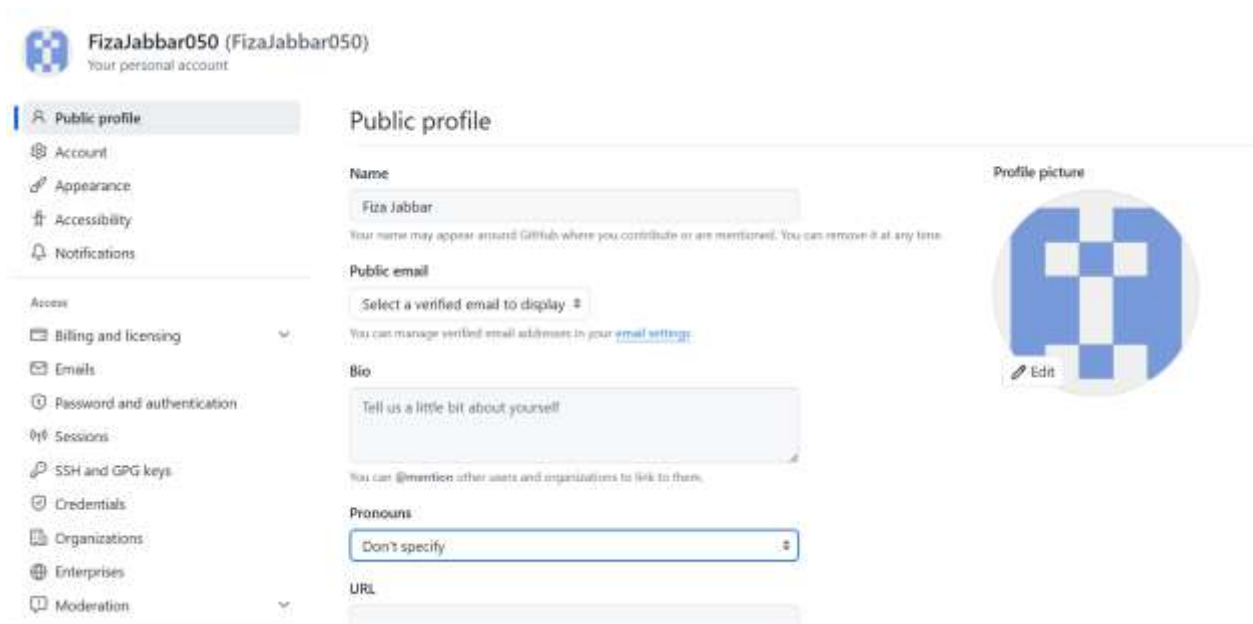
Section: VB

GitHub username : FizaJabbar050

Task 1: Create and Configure Your GitHub Account

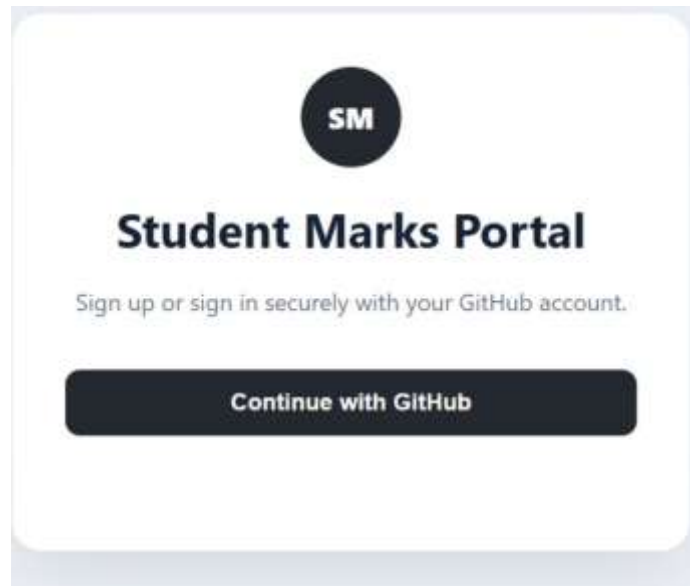


This screenshot shows the GitHub profile with the required username and profile information

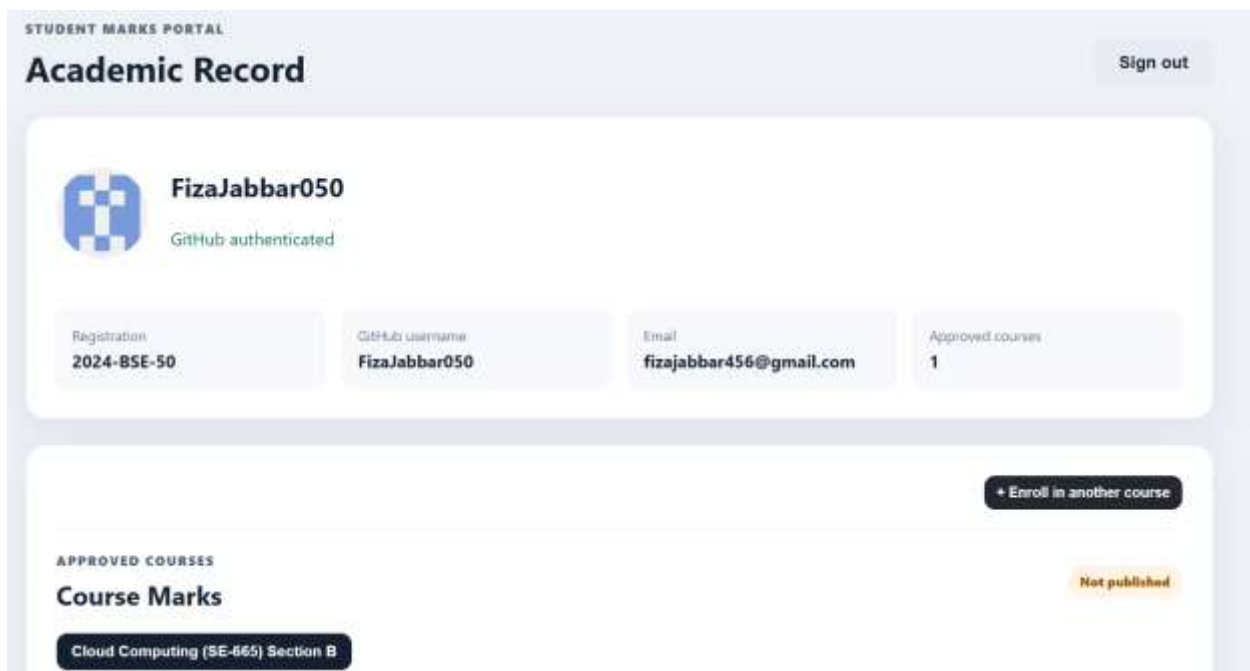


This screenshot shows the actual name entered in the GitHub profile.

Task 2: Register in the Student Marks Portal



This screenshot shows the Student Marks Portal login page.

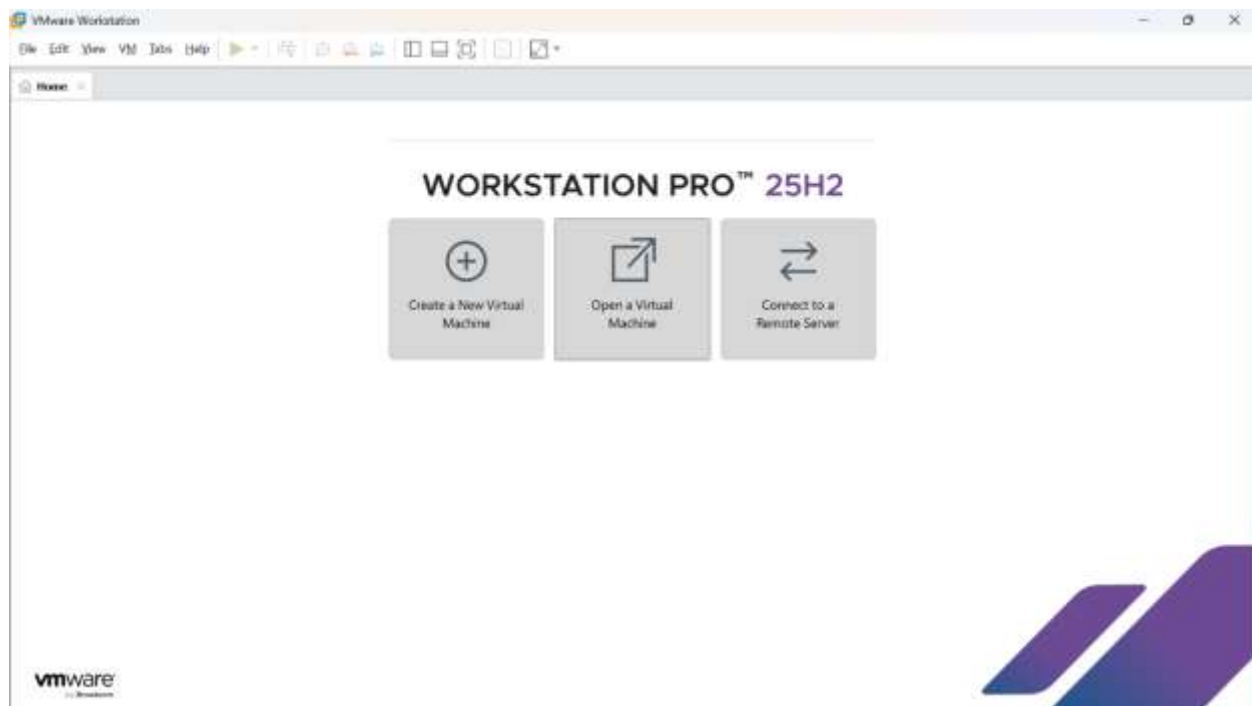


This screenshot shows the GitHub profile information linked to the portal



This screenshot shows the submitted enrollment and pending approval status.

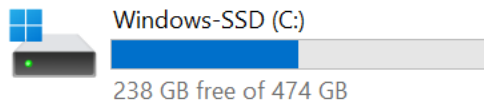
Task 3: Download VMware Workstation Pro and Ubuntu Server



This screenshot shows the Ubuntu Server ISO selected for the virtual machine.

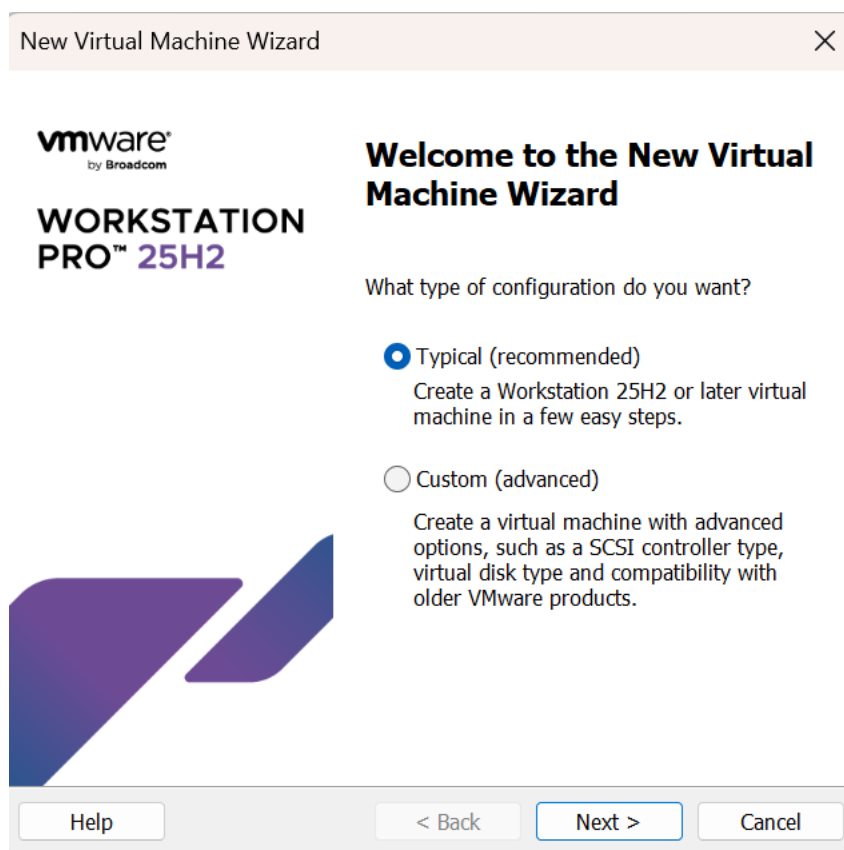
This screenshot shows the downloaded Ubuntu Server ISO file

Devices and drives



This screenshot shows the available storage for the virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine

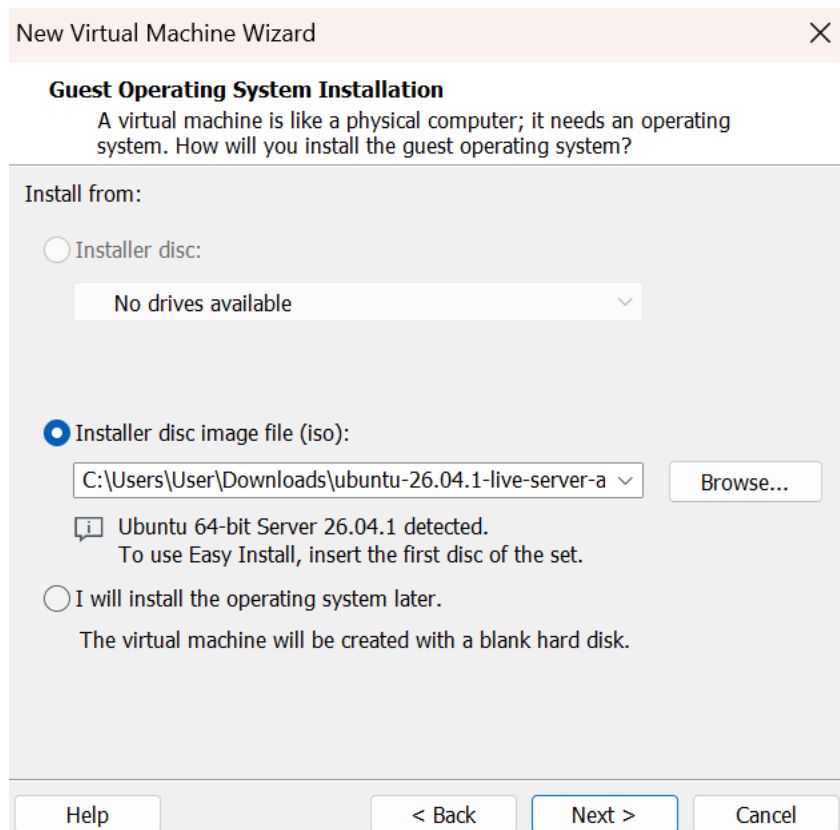


This screenshot shows the VMware virtual machine creation wizard.

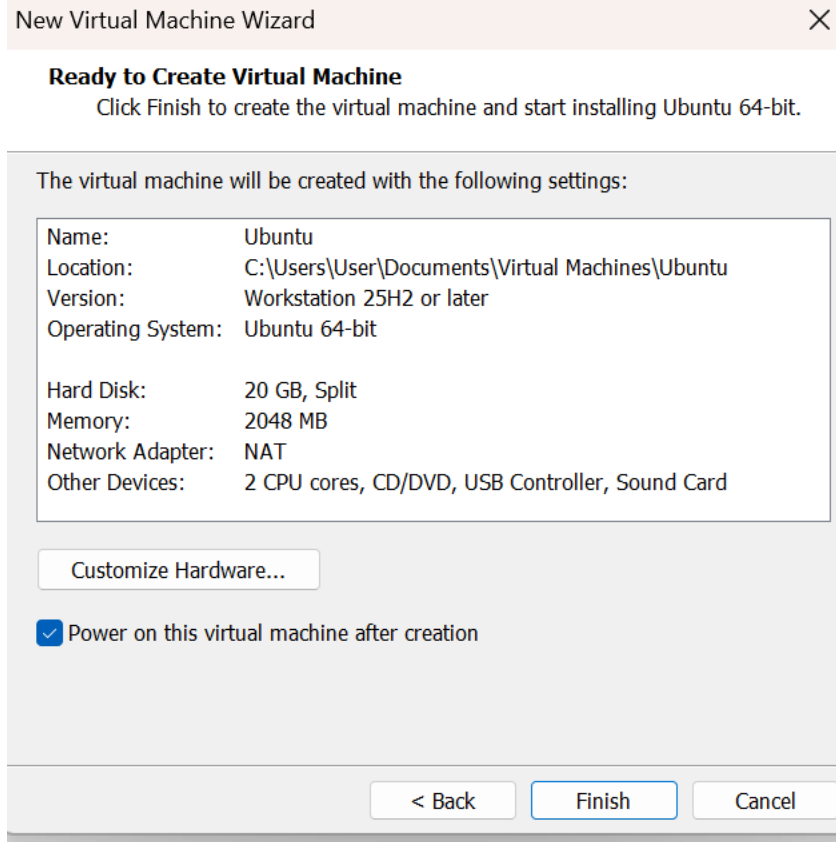
What type of configuration do you want?

- ☒ Typical (recommended)
Create a Workstation 25H2 or later virtual machine in a few easy steps.
- ☐ Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.

This screenshot shows the Typical configuration selected.



This screenshot shows the Ubuntu ISO selected in VMware



This screenshot shows the final virtual machine configuration summary

```
[ OK ] Finished systemd-tmpfiles-setup-dev: Create Static Device Nodes in /dev.
[ OK ] Reached target local-fs-pre.target: Preparation for Local File Systems.
[ OK ] Reached target local-fs.target - Local File Systems.
[ OK ] Started systemd-journald.service - Journal Service.
[ 2.876121] dracut-cmdline[174]: Warning: Kernel command line option 'copyhd
' is deprecated, use 'rd.driver.export' instead.
Starting systemd-tmpfiles-setup.service: Create System Files and Directories...
[ OK ] Finished dracut-cmdline.service - dracut cmdline hook.
Starting dracut-pre-udev.service - dracut pre-udev hook...
[ OK ] Finished systemd-tmpfiles-setup.service: Create System Files and Directories.
[ OK ] Finished dracut-pre-udev.service - dracut pre-udev hook.
Starting systemd-udevd.service - Manager for Device Events and Files...
[ OK ] Started systemd-udevd.service - Manager for Device Events and Files.
Starting systemd-networkd.service - Network Management...
Starting systemd-udev-trigger.service - Coldplug All udev Devices...
[ OK ] Started systemd-networkd.service - Network Management.
[ OK ] Reached target network.target - Network.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Reached target sysinit.target - System Initialization.
[ OK ] Reached target basic.target - Basic System.
Starting dracut-initqueue.service - dracut initqueue hook...
[ OK ] Created slice system-modprobe.slice - Slice /system/modprobe.
Mounting sys-kernel-config.mount - Kernel Configuration File System...
[ OK ] Mounted sys-kernel-config.mount - Kernel Configuration File System.
```

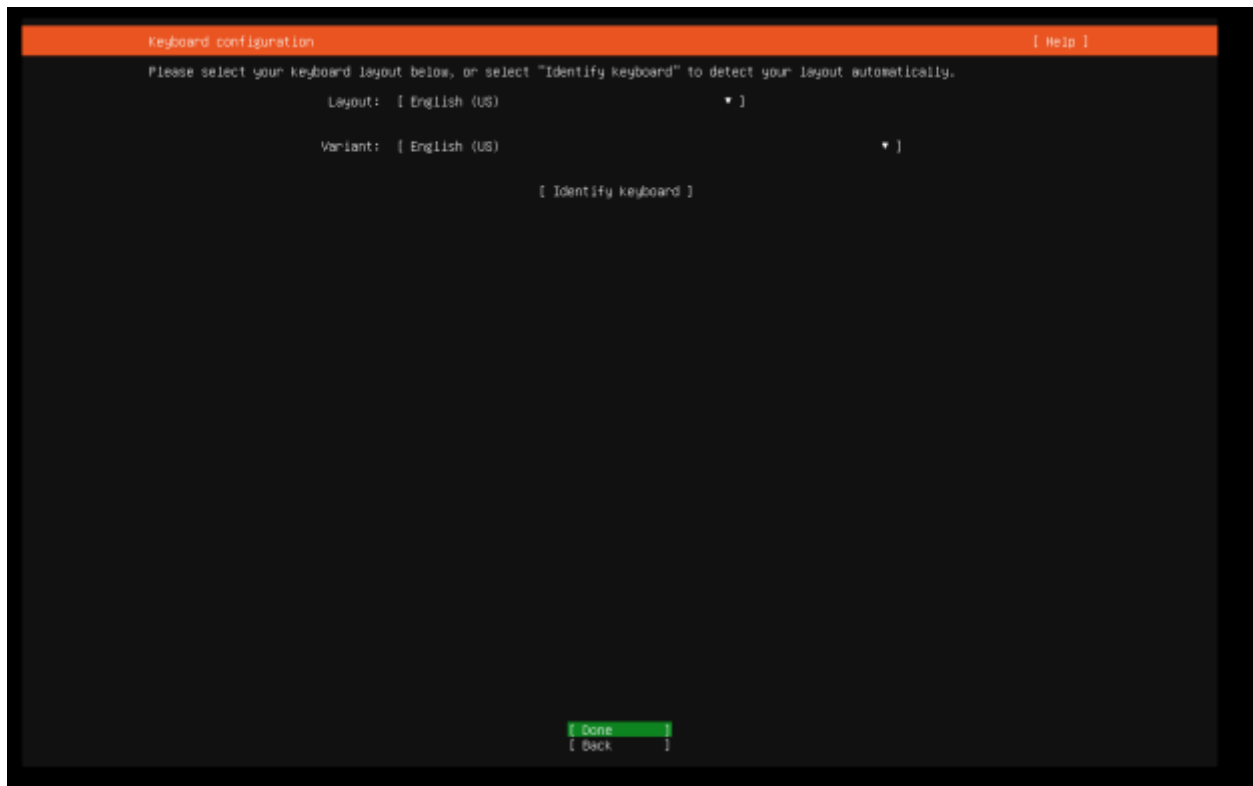
This screenshot shows the Ubuntu Server installer boot screen

Task 5: Choose the Installation Language

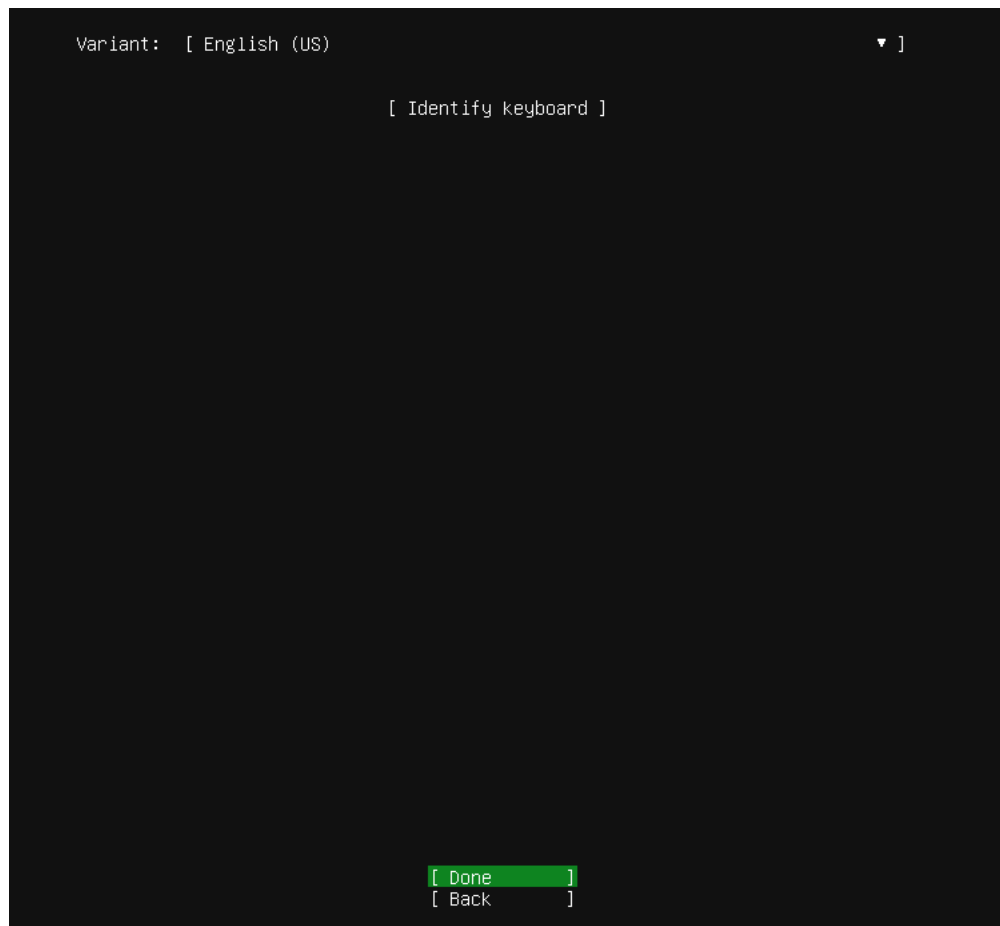


This screenshot shows English selected as the installation language

Task 6: Configure the Keyboard Layout



This screenshot shows the selected keyboard layout

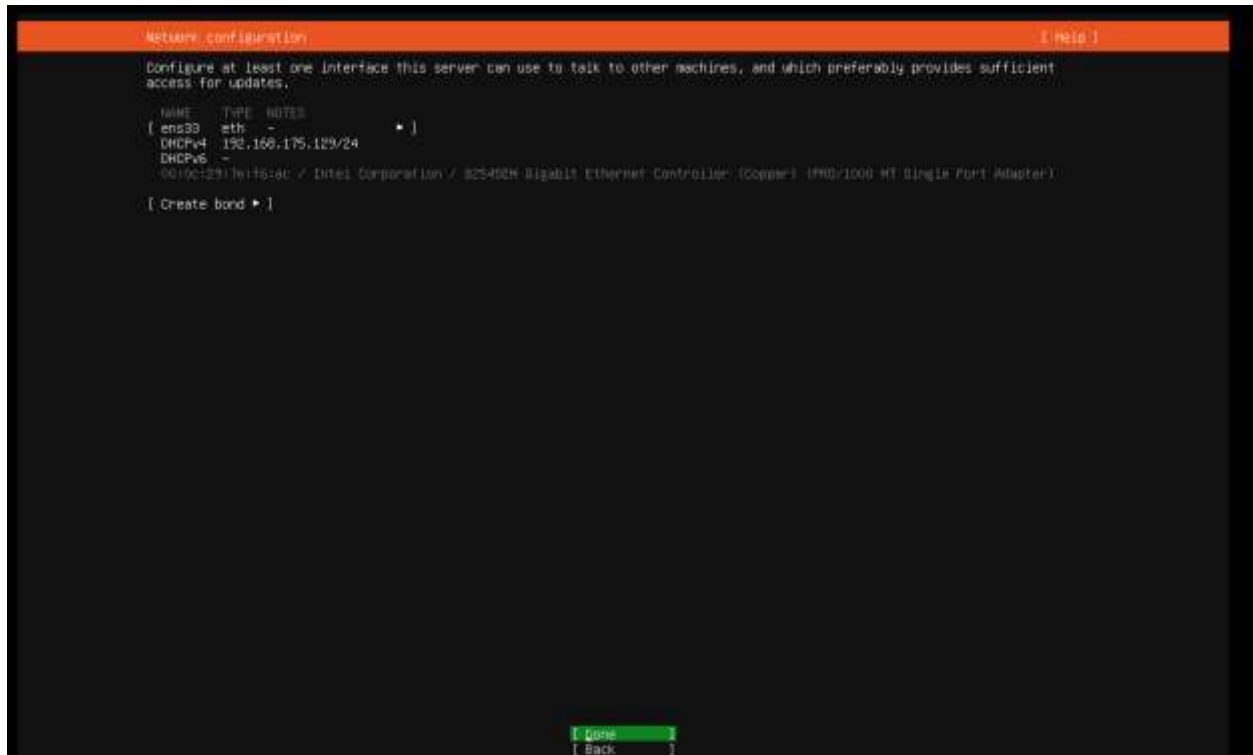


This screenshot shows the selected keyboard variant

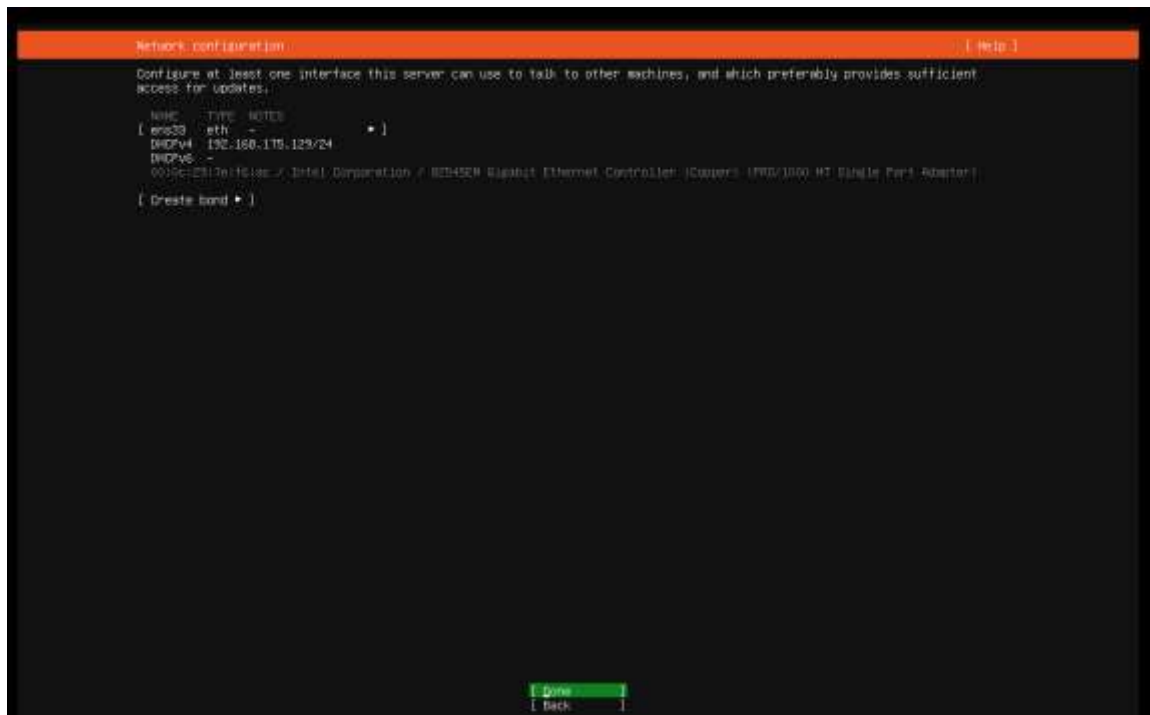
Task 7: Choose the Ubuntu Server Installation Type

No separate screenshot was captured because this step was merged by the Ubuntu installer

Task 8: Configure the Network

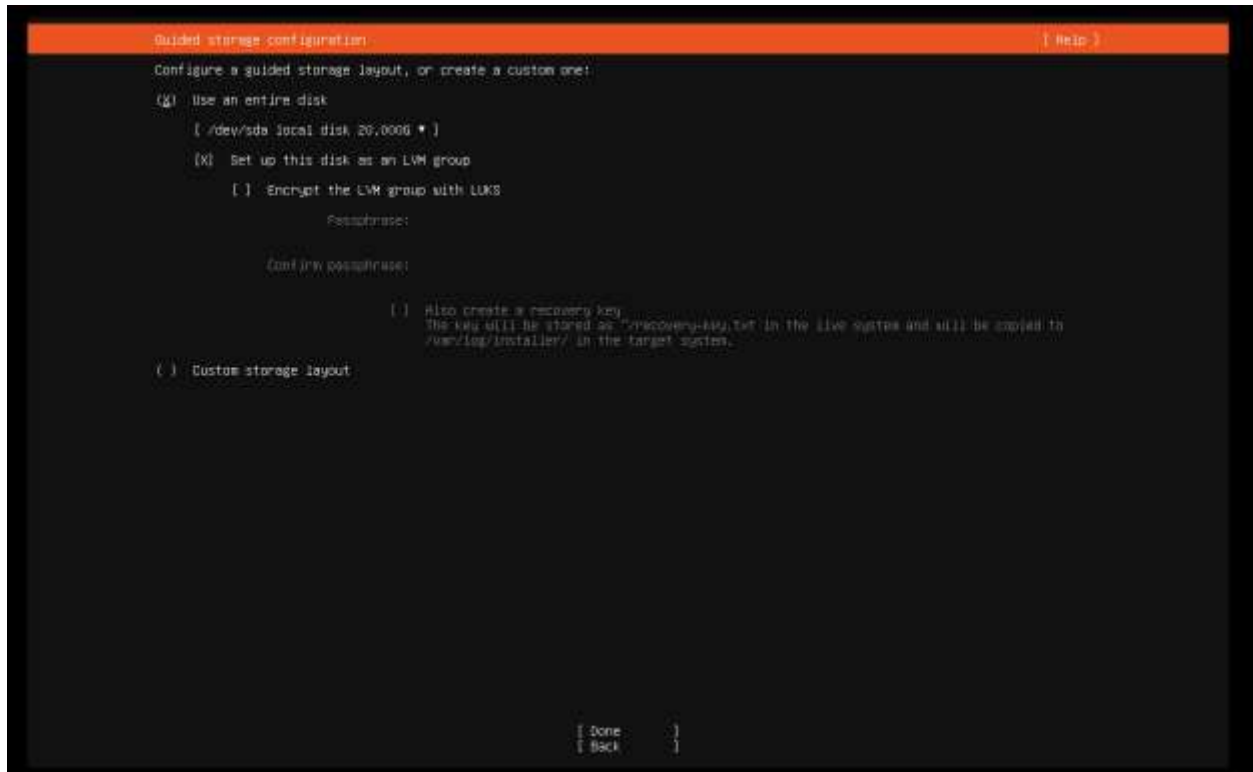


This screenshot shows the active network interface detected by Ubuntu

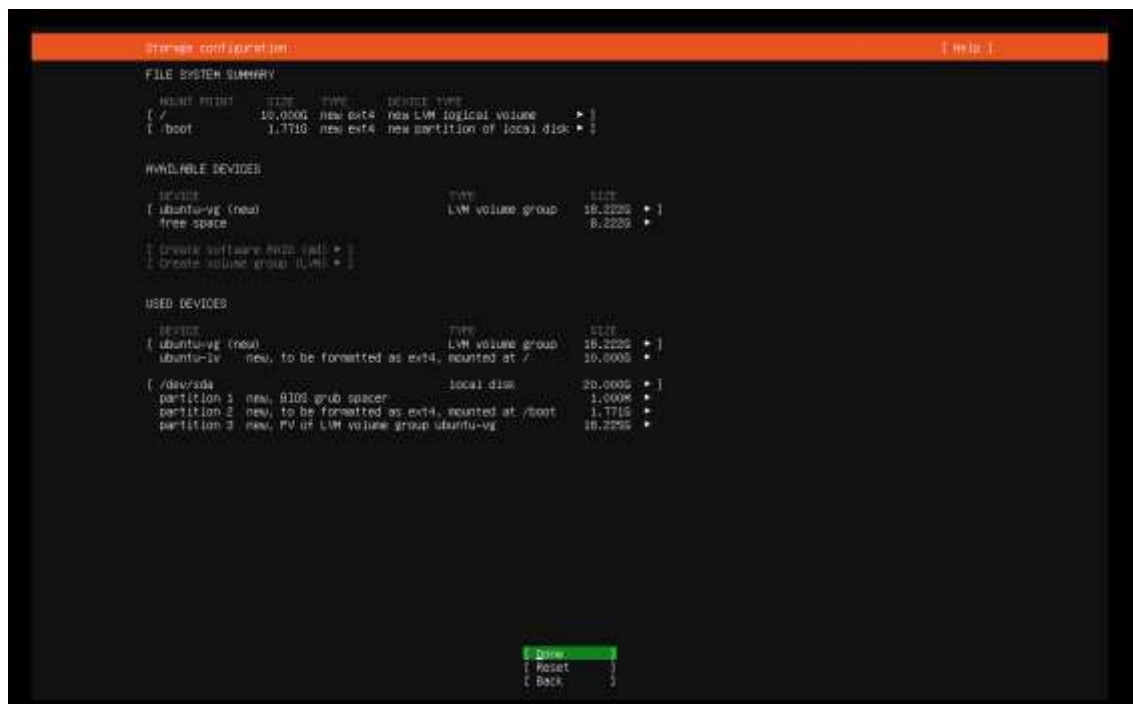


This screenshot shows the Ubuntu network configuration and IP address

Task 9: Configure Storage



This screenshot shows the Guided Storage configuration using the virtual disk



This screenshot shows the configured partitions and LVM storage

Task 10: Select the Installation Disk

```
DEVICE                                     TYPE      SIZE
[ /dev/sda                                local disk 20.000G ▶ ]
```

This screenshot showing the selected disk

```
USED DEVICES

DEVICE                                     TYPE      SIZE
[ /dev/sda                                local disk 20.000G ▶ ]
partition 1 new, BIOS grub spacer          1.000M ▶ ]
partition 2 new, to be formatted as ext4, mounted at / 19.997G ▶ ]
```

This screenshot showing the disk name and size

Task 11: Review the Partitions

```
FILE SYSTEM SUMMARY

MOUNT POINT  SIZE  TYPE  DEVICE TYPE
[ /           10.000G new ext4 new LVM logical volume ▶ ]
[ /boot       1.771G new ext4 new partition of local disk ▶ ]

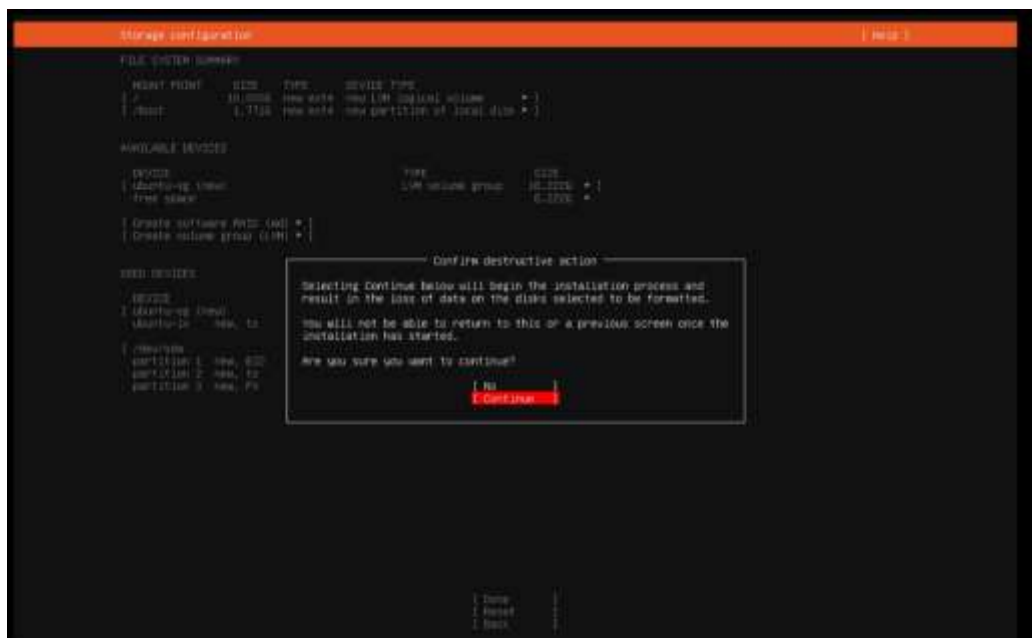
AVAILABLE DEVICES

DEVICE                                     TYPE      SIZE
[ ubuntu-vg (new)                          LVM volume group 18.222G ▶ ]
free space                                   8.222G ▶ ]

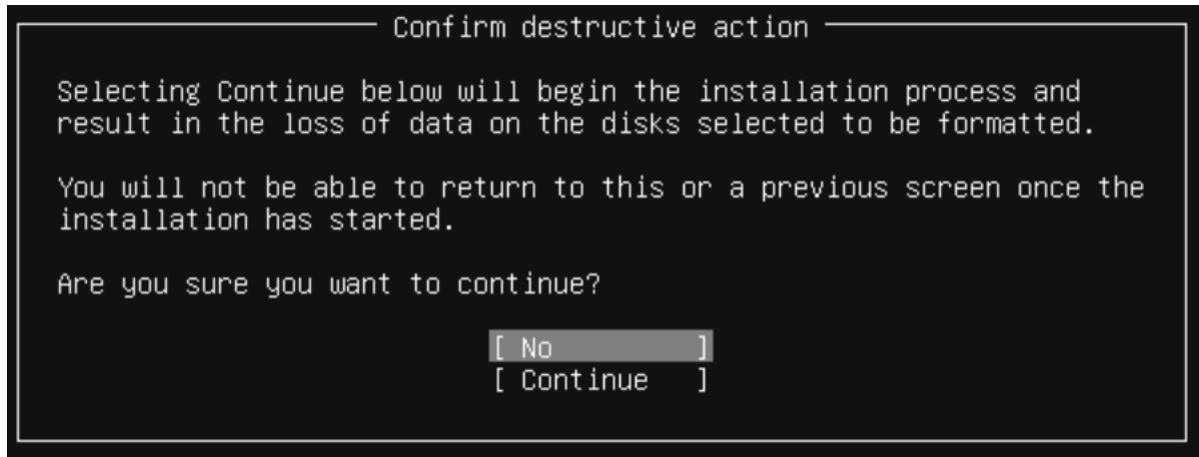
[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]
```

This screenshot showing the proposed partitions

Task 12: Confirm the Storage Changes



This screenshot showing the final warning



This screenshot showing the confirmation selection



This screenshot shows the actual name entered during Ubuntu setup



This screenshot shows the server name configured as ubuntu



This screenshot shows the Ubuntu username matching the GitHub username

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Fiza Jabbar

Your server's name:

ubuntu

The name it uses when it talks to other computers.

Pick a username:

FizaJabbar050

Choose a password:

Confirm your password:

[Done]

This screenshot shows the completed Ubuntu profile configuration

Task 14: Complete the Software Installation

Updating system

[Help]

```
configuring partition: partition-2
configuring format: format-0
configuring luksvolgroup: luksvolgroup-0
configuring lukspartition: lukspartition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install:
writing install sources to disk
running 'curtin extract'
curtin command extract:
acquiring and extracting image from cpi:///tftp/tftp1a12bctq/mount
configuring keyboard
curtin command in-target:
executing curtin install curthooks step
curtin command install:
configuring installed system
running 'curtin curthooks'
curtin command curthooks:
configuring apt configuring apt
installing missing packages
configuring local service
configuring raid (mdadm) service
configuring Wi-Fi over TCP
installing kernel
setting up sleep
apply networking config
writing etc/fstab
configuring multihost
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target:
```

[View full log]

This screenshot shows the Ubuntu Server installation in progress

Task 15: Reboot and Verify the Ubuntu Identity

```
Installation complete! [ Help ]

curtin command extract
  acquiring and extracting image from cp:///tmp/tmp4ap0hjic/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  applying networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
curtin command in-target
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
curtin command system-install
  unpacking openssh-server
curtin command system-install
  configuring cloud-init
  downloading and installing security updates
curtin command in-target
  restoring apt configuration
curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

This screenshot shows that the Ubuntu installation has completed

```
[ 33.308344] cloud-init[1141]: ..0.0 0 .
[ 33.507421] cloud-init[1141]: |..+*k.0 0 .
[ 33.508940] cloud-init[1141]: |.+*==+.+00 . ..
[ 33.510742] cloud-init[1141]: |00*B0 00E+= ..0.
[ 33.512660] cloud-init[1141]: +-----[SHA256]-----+

ubuntu login: _
```

This screenshot shows the Ubuntu Server login screen.

```
ubuntu login: fiza@bbr05a
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 7.0.0-51-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/support

System information as of Sun Sep 13 04:16:30 PM UTC 2020

System load:  0.02          Processes:      215
Usage of /:   20.5% of 9.75GB    Users logged in: 0
Memory usage: 11%             IPv4 address for ens33: 192.168.175.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

fiza@bbr05a:~$
```

This screenshot shows the user logged into the Ubuntu terminal

```
fizajabbar050@ubuntu:~$
```

This screenshot verifies the required Ubuntu username and server name

Task 16: Find the Ubuntu Server IP Address

```
fizajabbar050@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:7e:f6:ac brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c297ef6ac
    inet 192.168.175.130/24 metric 100 brd 192.168.175.255 scope global dynamic ens33
        valid_lft 1739sec preferred_lft 1739sec
    inet6 fe80::20c:29ff:fe7e:f6ac/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
fizajabbar050@ubuntu:~$
```

This screenshot shows the ip addr command and its output

```
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:7e:f6:ac brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c297ef6ac
    inet 192.168.175.130/24 metric 100 brd 192.168.175.255 scope global dynamic ens33
        valid_lft 1739sec preferred_lft 1739sec
    inet6 fe80::20c:29ff:fe7e:f6ac/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
fizajabbar050@ubuntu:~$ _
```

This screenshot clearly shows the active IPv4 address of the Ubuntu Server

Task 17: Connect to Ubuntu Server from Windows

```
C:\Windows\System32>ssh fizajabbar050@192.168.175.130
The authenticity of host '192.168.175.130 (192.168.175.130)' can't be established.
ED25519 key fingerprint is SHA256:1XR5nPPk87qsyBqJ+XgviyDswrWi31HKi4clPo2zZ/0.
This key is not known by any other names.
```

This screenshot shows the SSH command used from Windows.

```
ED25519 key fingerprint is SHA256:1XR5nPPk87qsyBqJ+XgviyDswrWi31HKi4clPo2zZ/0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.175.130' (ED25519) to the list of known hosts.
Connection closed by 192.168.175.130 port 22
```

This screenshot shows the server fingerprint being accepted

```
C:\Windows\System32>
C:\Windows\System32>ssh fizajabbar050@192.168.175.130
fizajabbar050@192.168.175.130's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep 14 12:50:12 PM UTC 2026

System load:  0.08               Processes:            222
Usage of /:   47.6% of 9.75GB    Users logged in:     0
Memory usage: 26%               IPv4 address for ens33: 192.168.175.130
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

This screenshot shows the successful SSH login.

```
fizajabbar050@ubuntu:~$
fizajabbar050@ubuntu:~$
```

This screenshot verifies the required remote Ubuntu prompt

Commands

```
sudo apt update
```

```
sudo apt install -y openssh-server
```

```
sudo systemctl enable --now ssh
```